RYO-CHAN Hackathon 2026 — Project Submission Form

Team: **Jason Parser Security** · Project: **Decision Receipt** · Track: **Autonomous Agents**

Members

Total Members	1
Name	casey_blanche
Role	Lead builder / agent design
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Discord / Telegram	Node0 (node0_68581) · Server Tag AGNS
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Overview

Project	Decision Receipt — advise-only research agent that turns market evidence into a shareable Decision Receipt (evidence → Bull/Bear debate → judgment → practice plan) for RealCryptoCap AI chat users (Free / Pro / Ultra).
Track	Track 1 — Autonomous Agents
Problem	Traders and holders ask “should I consider this asset?” but get either a single opaque model reply or a wall of charts. They cannot see which evidence drove the call, what the opposing case is, or what would change the read. Existing AI chat tools rarely combine multiple research sources into one auditable decision.
Solution	A research agent on RYO’s read-only MCP layer plus RealCryptoCap live coin/market data. It (1) checks RealCryptoCap universe coverage with positive messaging when an asset is out of scope, (2) runs the RYO research funnel, (3) runs Bull and Bear agents on the same evidence pack, (4) has a Judge issue stance + confidence, (5) emits a Decision Receipt with a practice plan only — never an order. Failures (rate limits, missing key, partial data) are first-class notes in the receipt.
Key Features	• Coverage gate with positive “not listed yet / by design” messaging for stablecoins, wrapped/staked, tokenized stocks/funds/treasuries/RWA, commodities • Multi-source evidence (RYO tools + RealCryptoCap market/coin/chart) • Bull/Bear debate from the same structured evidence • Judge stance + confidence • Practice plan (entry idea, invalidation, size hint) — not executed • Shareable JSON + text Decision Receipt • Graceful degradation when RYO key missing or a tool is partial/unavailable
Target Users	RealCryptoCap AI chat users on Free, Pro, and Ultra tiers who want explainable research before any decision. Also builders/judges who want to audit the reasoning path.
Scope	Read-only research + practice-plan generation + shareable receipts. Integrates RYO MCP research tools and RealCryptoCap public/Pro market endpoints. Optional Tavily news fusion when a theme needs news. In-product tier limits: Free limited, Pro higher limits, Ultra highest.
Limitations	• No trading, custody, or order routing (RYO tools are read-only by design) • Not financial advice; practice plan only • RYO MCP key must be issued by organizer (placeholder-ready) • Assets outside RealCryptoCap’s native universe are intentionally not receipted • Demo uses deterministic debate; LLM polish optional when LLM keys are set

Tech Stack

Frontend Stack	Receipt text/JSON output designed for RealCryptoCap AI chat surfaces (CLI first for hackathon demo; chat tool-calling ready).
Backend Stack	Node.js 20+ (ESM). REST clients for RYO MCP (https://app-ryochan.com/api/mcp) and RealCryptoCap APIs (https://realcryptocap.com). Soft retry + rate-limit headers.
AI Model(s)	Bring-your-own model via env (LLM_BASE_URL / LLM_API_KEY / LLM_MODEL). Deterministic heuristic debate ships default so the agent is repeatable without an LLM. Compatible with RealCryptoCap Free/Pro/Ultra model routing.
Other Technologies	RealCryptoCap asset universe filter (native crypto only), public JSON market-data/coin/chart endpoints, optional Tavily Researcher (1k credits/mo free plan). Secrets via .env.example only.

Repository / Demo

GitHub Repository	Private RYO organizer repo (pending invite). Local package: **decision-receipt**. Public snapshot policy: jamesparser / redacted only — no proprietary RealCryptoCap API/MCP/CLI/market-data internals.
Demo Video	TBD — record after RYO_MCP_KEY is filled (see Testing).
Documentation	README.md · .env.example · MCP Builder Guide (organizer) · this form.

Testing Information

How to run the Project	cp .env.example .env → set RYO_MCP_KEY → npm start -- demo SOL
Prerequisites	Node.js 20+, network access to app-ryochan.com and realcryptocap.com.
Installation Steps	1) unzip/clone package 2) cp .env.example .env 3) paste RYO_MCP_KEY from organizer DM 4) node --test test/receipt.test.js 5) node src/cli.js demo SOL
Environment Variables	RYO_MCP_URL, RYO_MCP_KEY (required), RCC_BASE_URL, RCC_API_KEY (optional Pro/Ultra), TAVILY_API_KEY (optional), LLM_* (optional), DECISION_RECEIPT_OUT_DIR
Build Command	None required (pure Node ESM).
Run Command	node src/cli.js demo SOL · node src/cli.js coverage USDT · node src/cli.js health

Test Command

node --test test/receipt.test.js · npm test
Smoke: node src/cli.js health · node src/cli.js coverage USDT · node src/cli.js demo SOL

Test Account(s)

None required for core receipt flow (public RealCryptoCap data + RYO MCP builder key). Optional: RealCryptoCap Pro/Ultra API key for higher limits; Tavily key for news themes. Demo uses the builder's own practice-plan state only.

Social Media Award (X)

Team X: **@JasonParserSec** — will post a public BUIDL update tagging **@ryodigital**, show what we built and why it matters, and include the demo/BUIDL link. Post URL will be added to the web submission before judging begins.

Notes for judges

• Advise-only: RYO tools cannot trade; practice plans are never executed.
• RealCryptoCap does not track stablecoins, tokenized stocks/funds/treasuries/RWAs, wrapped/staked receipts, or commodities — coverage messages are positive and product-true.
• Combining sources + coping with failure are first-class (envelope status, failure notes, missing-key fallback).